WAYAMBA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

Level III Semester I – (Academic Year 2023/2024)

Department of Computing and Information Systems

CMIS 3142 – Computational Methods

Assignment 06

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1. Consider the following system of linear equations:

   $10x+y+z = 12$

   $2x+10y+z = 13$

   $2x+2y+10z = 14$

   a. Derive the iterative equations for x, y, and z using the Jacobi method.
   b. Starting with the initial approximation $x^{(0)} = y^{(0)} = z^{(0)} = 0$, calculate the first three iterations. Show your calculations in a table.
   c. Write a Java method to implement the Jacobi method. The method should accept the initial values, tolerance, and maximum number of iterations as parameters and return the approximate solution.

2. Consider the following system of linear equations:

   $8x+y-z = 8$

   $2x+10y+z = 13$

   $x-y+6z = 6$

   a. Rearrange the equations to obtain the iterative formulae for the Gauss–Seidel method.
   b. Using $x^{(0)} = y^{(0)} = z^{(0)} = 0$, calculate the first three iterations.
   c. Write a C or Java program to solve the system using the Gauss–Seidel method. Terminate the iteration when the maximum absolute change between consecutive approximations is less than 10^{-4}, or when 100 iterations have been completed.

3. Consider the diagonally dominant system:

$$10x - y + 2z = 6$$
$$-x + 11y - z = 22$$
$$2x - y + 10z = -10$$

a. Determine whether the system satisfies the strict diagonal dominance condition. Explain how this condition relates to the convergence of iterative methods.

b. Perform two iterations of the Jacobi method and three iterations of the Gauss–Seidel method, using the initial approximation (0,0,0). Present the results in a comparison table.

c. Write a Java program that implements both methods and displays the approximate values of x, y, and z after each iteration.

Submit on or before: 18 – Oct – 2026